

Overview

Language models for biology

Transformer Language Models pretrained on 100M+ sequences learn functional patterns beyond simple statistics.

Central question

Do large pretrained LMs consistently outperform engineered baselines - or is their advantage task-dependent?

DNA Task – Promoter Classification

Binary classification · 200 bp sequences · Human GRCh38
4,000 sequences · 2,000 promoter / 2,000 non-promoter ·
Balanced split

Protein Task – Pfam Classification

10-class classification · UniProt Swiss-Prot · 10 Pfam families

2,293 sequences · GPCRs, kinases, GTPases, TFs, zinc fingers, immune & ankyrin domains

Methodology

Raw Sequences

GRCh38 DNA (200 bp) | UniProt Protein

Feature Engineering

95 DNA features | 176 Protein features

Transformer Embeddings

DNABERT-2 / NT-500M | ESM-2 / ProtBERT

Downstream Classifiers

LR · SVM · Random Forest · XGBoost

Evaluation & Comparison

ROC-AUC (DNA) | Accuracy (Protein)

Data Sources

- ▶ DNA: GRCh38 genome · 200 bp promoter/non-promoter
- ▶ Protein: UniProt Swiss-Prot · 10 Pfam families

Feature Engineering

- ▶ DNA: 95 features — GC content, CpG O/E, k-mer freq, Shannon entropy
- ▶ Protein: 176 features — CTD, PseAAC, reduced dipeptides

Modeling Paradigms

- ▶ Baseline ML · Sequence CNN · Hybrid (CNN + features)
- ▶ Transformer LMs: DNABERT-2, NT-500M, ESM-2, ProtBERT

Transformer Models

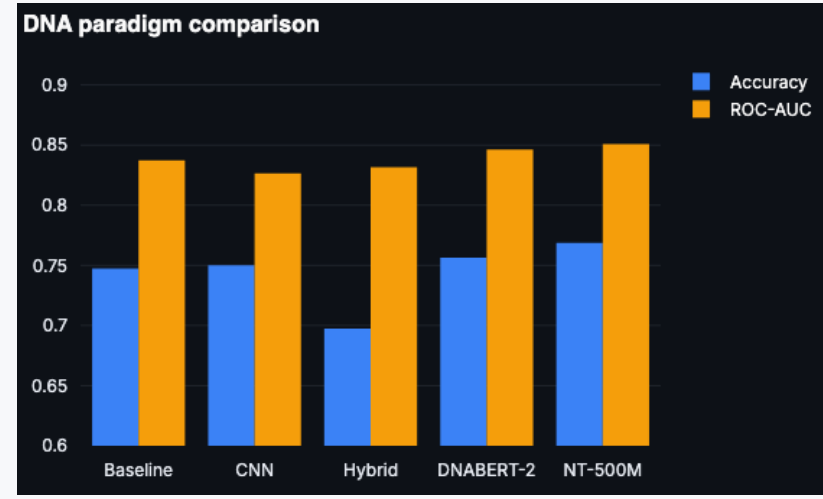
- ▶ DNABERT-2: 117M params · BPE · 135-species genomes
- ▶ NT-500M: 500M params · GRCh38 human genome
- ▶ ESM-2: 35M params · 65M UR50D proteins
- ▶ ProtBERT: 420M params · 217M UniRef100

Evaluation & Infrastructure

- ▶ DNA: ROC-AUC · Protein: Accuracy + F1-macro
- ▶ 5-fold CV · Stratified 80/20 train/test split
- ▶ UF HiPerGator HPC · NVIDIA A100 GPUs · SLURM

Results

Paradigm	Best Model	Accuracy	ROC-AUC
Baseline ML	Random Forest	0.748	0.837
CNN	Seq_Cnn	0.750	0.826
Hybrid	CNN + features	0.698	0.832
DNABERT-2	XGBoost	0.756	0.846
NT-500M ★	XGBoost	0.769	0.851



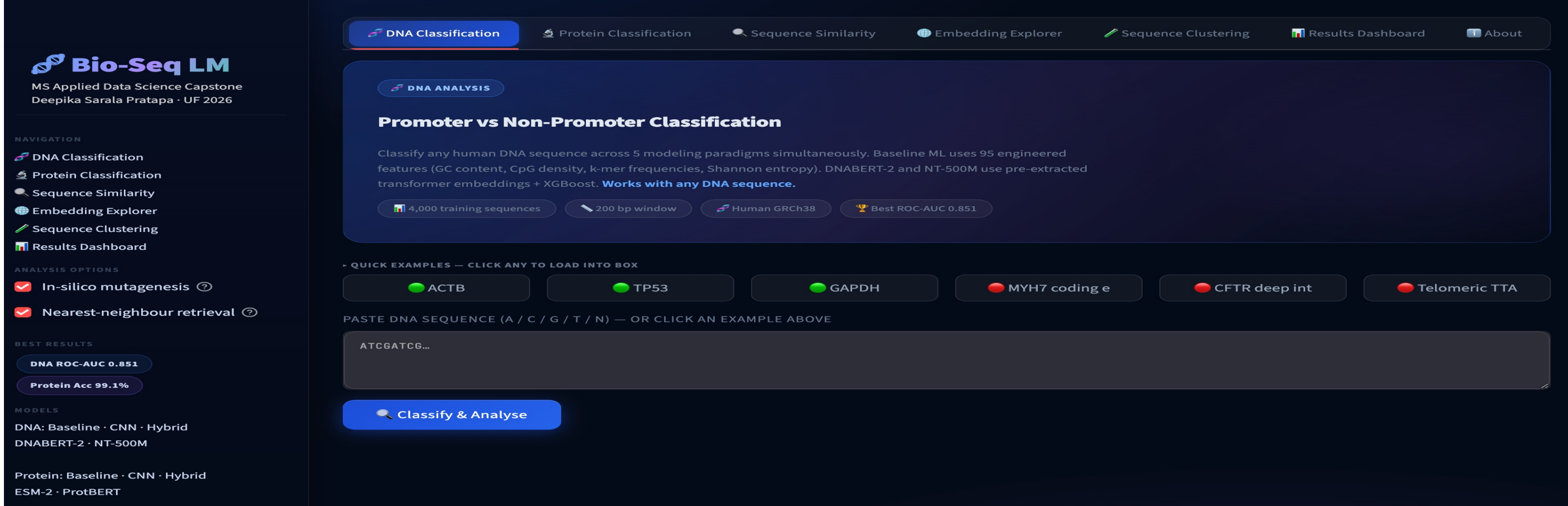
NT-500M / XGBoost achieves ROC-AUC 0.851 — only +1.5 pts above Random Forest baseline. Engineered CpG/GC features remain highly competitive.

Paradigm	Best Model	Accuracy	F1-macro
Baseline ML	Random Forest	0.923	0.924
CNN	seq_cnn	0.924	0.925
Hybrid	CNN + features	0.956	0.957
ESM-2	Linear SVM	0.987	0.989
ProtBERT ★	Linear SVM	0.991	0.992



ProtBERT / Linear SVM achieves 99.1% accuracy — +7 pts over Hybrid. Pfam families are linearly separable in UniRef100 pretrained embedding space.

Interactive App – Bio Seq LM Explorer



Real-time biological sequence analysis with prediction, similarity search, clustering, and interpretability tools.

Embedding Space



Protein embeddings show clear clusters, while DNA embeddings exhibit overlap indicating stronger separability for protein tasks.

Interactive App – Live Predictions



Interpretability — Mutagenesis & Features

In-Silico Mutagenesis (DNA)

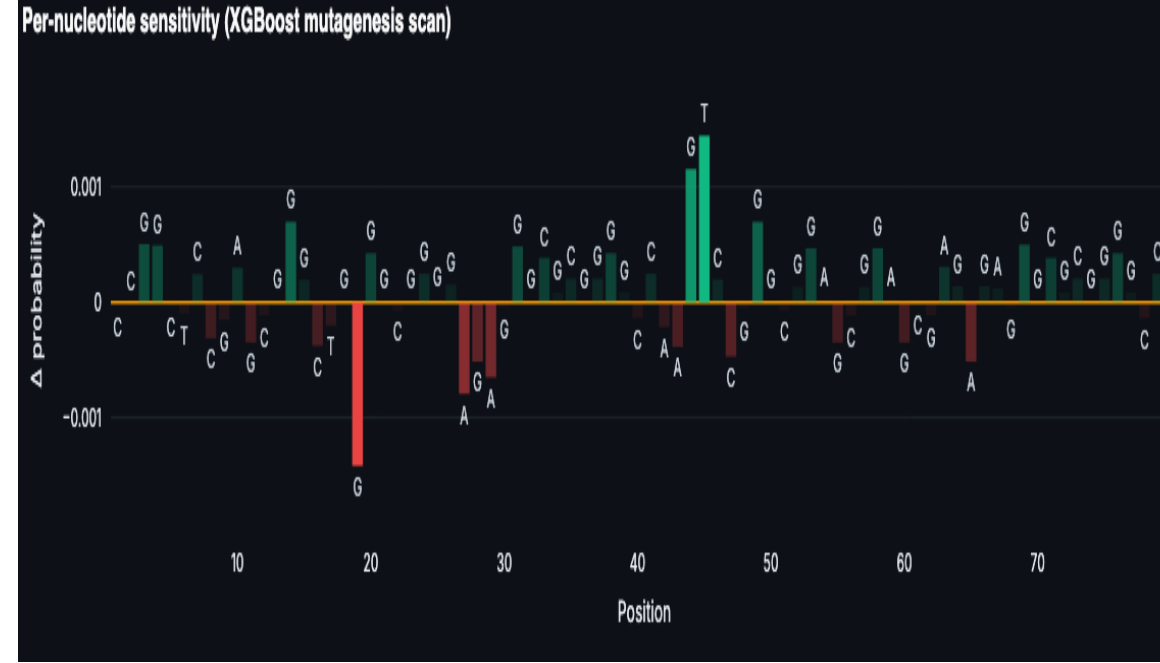
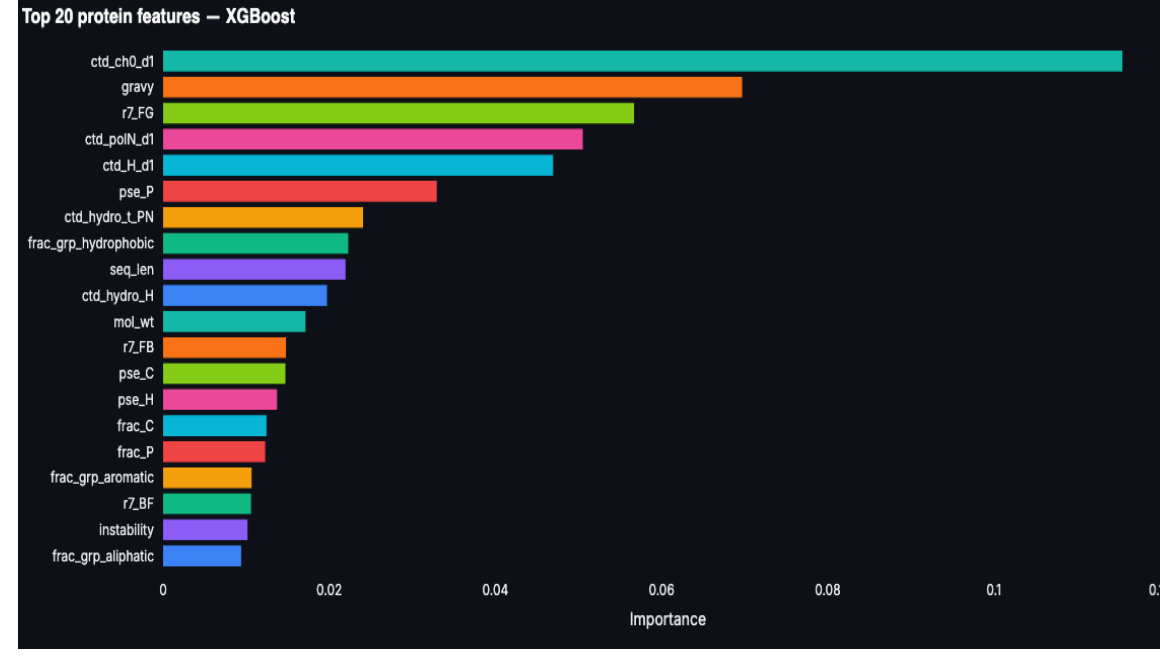
- ▶ Substitute each nucleotide with 3 alt bases
- ▶ Score $\Delta P(\text{promoter})$ per position
- ▶ Green = $\uparrow$ probability · Red = $\downarrow$ probability

Alanine Scanning (Protein)

- ▶ Residue substitution → functional site detection

Feature Importance (XGBoost)

- ▶ Top: GRAVY, CTD descriptors, reduced dipeptides
- ▶ Confirms biochemical signals drive classification



Key Takeaways

1. Transformer models significantly improve protein classification (+7% accuracy over hybrid models)
2. DNA classification remains driven by biological features (GC content, CpG density).
3. Protein embedding spaces are highly separable, enabling near-linear classification
4. Interpretability methods (mutagenesis, feature importance) reveal biologically meaningful signals
5. The unified pipeline enables analysis of unseen sequences without retraining

References

- [1] Ji et al. DNABERT-2 (2023). arXiv:2306.15006
- [2] Dalla-Torre et al. Nucleotide Transformer (2023). bioRxiv
- [3] Elnaggar et al. ProtBERT (2021). IEEE TPAMI
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- [5] Pedregosa et al. Scikit-learn (2011). JMLR